

The Sound of Your Secrets

Teaching Your Model to Spy, So You Can Learn to Defend

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David vonThenen

- Are you Human or an AI?
- I want 5 Kubernetes
- Virtual Machines are Real
- Cloudy, cloudy, cloudy...
- There is storage for that!

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Agenda

- **The Moral Duality Of AI / ML**
- **Build An Acoustical Keystroke Logger?**
 - **Dataset Processing Pipeline**
 - **Model Architecture, Enhancements**
 - **Live Demos! Yes, Plural!**
- **Real World Attack Vectors**
- **Privacy, Security, & Protecting Yourself**
- **Q&A**

The Moral Duality of AI / ML

Extreme Positive Outlook

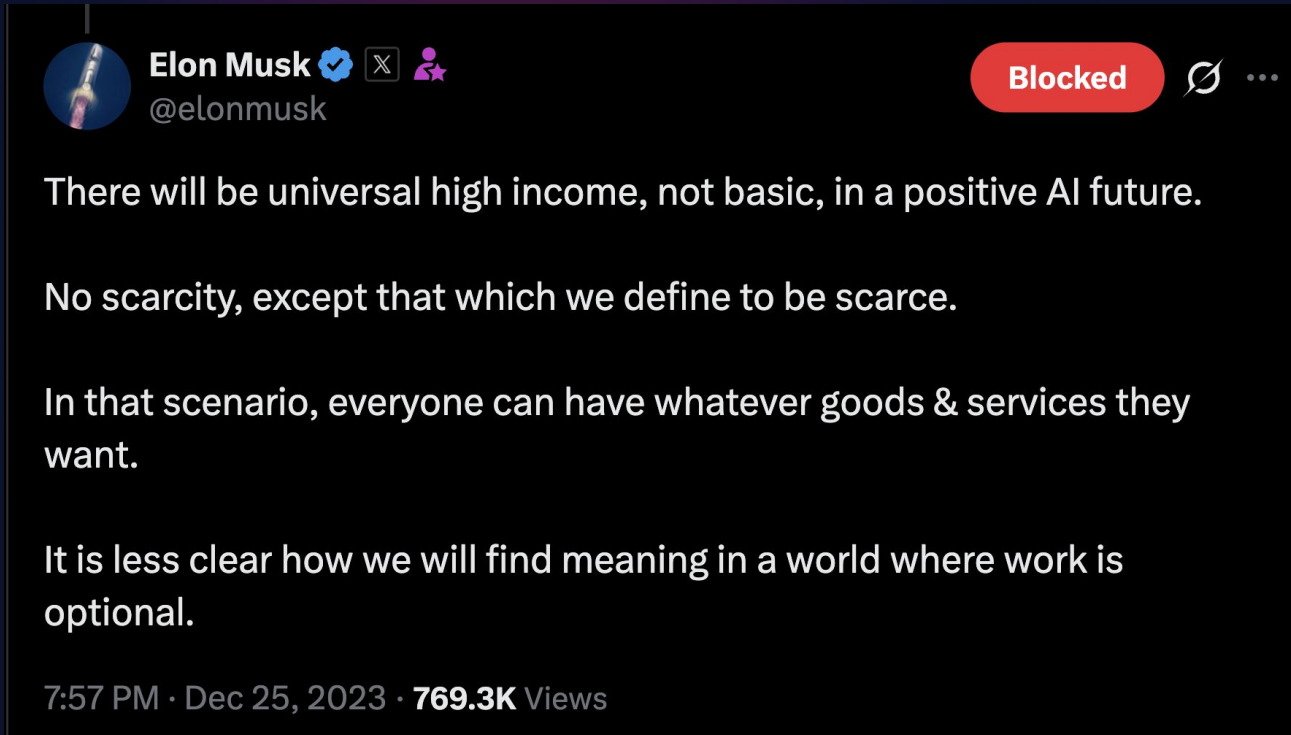


Image Attribution

Elon Musk - <https://x.com/elonmusk/status/1739495563043701011?s=20>

Extreme Negative Outlook

TECH

MIT study finds AI can already replace 11.7% of U.S. workforce

PUBLISHED WED, NOV 26 2025 10:00 AM EST | UPDAT TECH • AI



MacKenzie Sigalos
@KENZIESIGALOS

Employment for computer programmers in the U.S. has plummeted to its lowest level since 1980—years before the internet existed



By Sasha Rogelberg
Reporter

March 17, 2025, 2:00 PM ET

[Add us on](#) 

Image Attribution

<https://fortune.com/2025/03/17/computer-programming-jobs-lowest-1980-ai/>

<https://www.cnn.com/2025/11/26/mit-study-finds-ai-can-already-replace-11point7percent-of-us-workforce.html>

Curing All Disease

► *Pharmaceutics*. 2024 Oct 14;16(10):1328. doi: [10.3390/pharmaceutics16101328](https://doi.org/10.3390/pharmaceutics16101328) 

Artificial Intelligence (AI) Applications in Drug Discovery and Drug Delivery: Revolutionizing Personalized Medicine

[Dolores R Serrano](#)^{1,2,*}, [Francis C Luciano](#)¹, [Brayan J Anaya](#)¹, [Baris Ongoren](#)¹, [Aytug Kara](#)¹, [Gracia Molina](#)¹,
[Bianca I Ramirez](#)¹, [Sergio A Sánchez-Guirales](#)¹, [Jesus A Simon](#)¹, [Greta Tomietto](#)¹, [Chrysi Rapti](#)¹, [Helga K Ruiz](#)¹,
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Editor: Thierry Vandamme

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PMCID: PMC11510778 PMID: [39458657](https://pubmed.ncbi.nlm.nih.gov/39458657/)

Image Attribution

Artificial Intelligence (AI) Applications in Drug Discovery and Drug Delivery: Revolutionizing Personalized Medicine
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11510778/>

Maximum Money Extraction

ALGORITHMIC
PRICING

where companies hand pr



Instacart

We're excited to announce that we've acquired Eversight, an AI-powered promotions platform for CPG brands and retailers.

This Session...

arXiv > cs > arXiv:2308.01074

Computer Science > Cryptography and Security

[Submitted on 2 Aug 2023]

A Practical Deep Learning–Based Acoustic Side Channel Attack on Keyboards

Joshua Harrison, Ehsan Toreini, Maryam Mehrnezhad

With recent developments in deep learning, the ubiquity of micro-phones and the rise in online services via personal devices, acoustic side channel attacks have become a greater threat to keyboards than ever. This paper presents a practical implementation of a state-of-the-art deep learning model in order to detect keystrokes using a smartphone integrated microphone. When trained on keystrokes recorded by a nearby phone, the classifier achieved an accuracy of 95% when using a language model. When trained on keystrokes recorded using the video-conferencing software Zoom, an accuracy of 85% was seen without the use of a language model. Our results prove the practicality of these side channel attacks via off-the-shelf equipment and algorithms. We discuss methods to protect users against these series of attacks.

Image Attribution

A Practical Deep Learning-Based Acoustic Side Channel Attack on Keyboards

<https://arxiv.org/abs/2308.01074>

Building An Acoustical Keystroke Logger

ML Audio Classification?

- MANY Doing This!
- Spectrogram Comparison
 - Visual Representation
 - Similarities Visually
- Example Projects:
 - Cats vs Dogs
 - Environmental Sounds
 - Gunshot Recognition

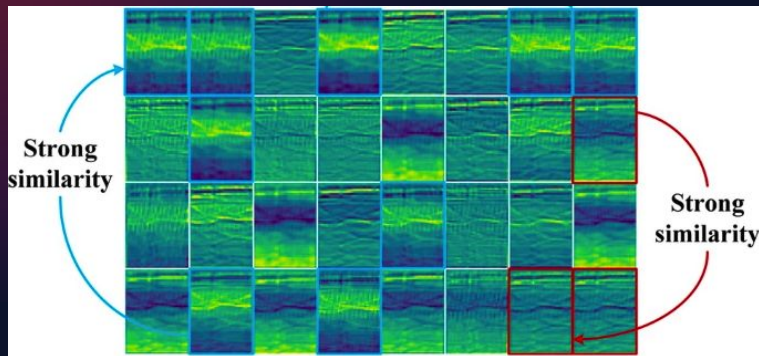
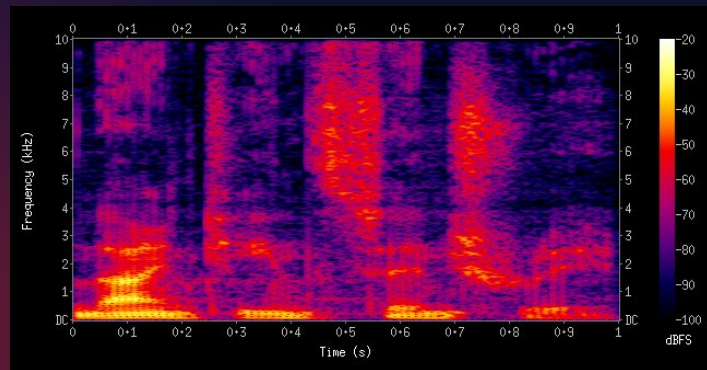


Image Credit:

[Fast environmental sound classification based on resource adaptive convolutional neural network](#) DOI:[10.1038/s41598-022-10382-x](#)

Sarcasm Detection

Video:

AIDev Europe: Sarcastically Speaking: Unlocking Multimodal Sentiment Analysis w/ NLP+Facial Landmark

<https://www.youtube.com/watch?v=9zxhjWNcCD8&list=PLbt-vfK614XogHSmJ0CNmUXtaUKYM247t&index=2>



Always Starts w/ The Dataset...

- Keystroke-noiseless-final dataset
 - "Sonotype" challenge at Singapore AI CTF 2024
 - Same problem...
- Multi-Keyboard Acoustic (MKA) Dataset
 - Previous Attempt At This
 - Persistent Inherent Limitations
 - Doesn't Encompass Complexity of Real-World Scenarios
- Will Provide Datasets In GitHub



Data Processing Pipeline

Recordings → Spectrogram Images

- Directory Name → Classification
- Labels (A-Z, 0-9, Sp, Ent) = 38
- Frequencies Over Time
- RGB (3 channels)
- Resized to a Fixed Grid Data Pipeline
- Normalization (Color + Brightness)
- Robust (Time Shift + Blank Mask)
- Train on "Features" Within The PNG



Model Training (skip)

CoAtNet-style Architecture

- Local Patterns, Global Context
- Loss = Cross Entropy
 - Multi-Class Classification
- Uncommon High Epochs (1200)
 - LR $5e-4$, Linear Annealing
 - Optimizer: AdamW (Adam + Weight Decay)
- Compute Train Loss + Train Accuracy Per Epoch



Acoustic Key Detection (Single)

- WAV for Training, Validation, Test Are All The Same!

Train



Validation



Test



Demo: Spectrographic Images

<https://youtu.be/Zn8xEkejHlQ>

Demo: Single Keyboard Train + Inference

<https://youtu.be/iD3GuFPcYN8>

Acoustic Key Detection (Multi)

Lenovo



Keystroke Dependent Dataset

Train

Lenovo



Validation

Lenovo



Test

Lenovo



Demo: Multi Keyboard Train + Inference

<https://youtu.be/1I4XW8JNnww>

Demo: Correcting Probabilities

<https://youtu.be/eXPG-IM3j5M>

Demo: Using AI To Enhance

https://youtu.be/_MG1q0g1-8w

Results From Both Experiments

Single Keyboard

- Accuracy: **100%**



Multiple Keyboards

- Accuracy (Exact): **46%**
- Even Distribution: 3%
 - Random: 1 out of 38

Additional Considerations:

- TopK > 1, Confid. Scores
- All Permutations
- AI to "Fill In The Blanks"

Keystroke Independent Dataset

Train

Lenovo



Validation

Lenovo

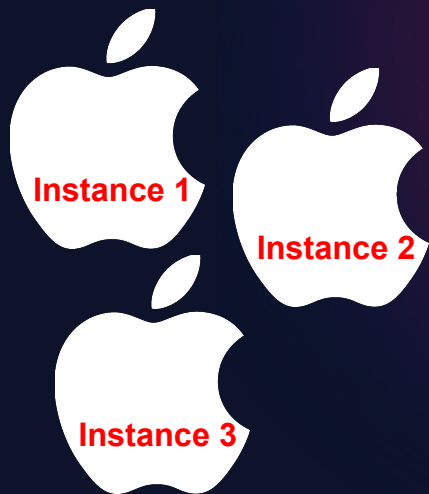


Test

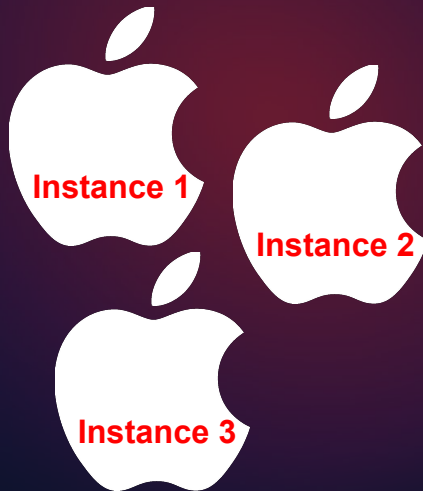


Manufacturing Similarity Dataset

Train



Validation



Test



Letter	Count	Letter	Frequency
E	21912	E	12.02
T	16587	T	9.10
A	14810	A	8.12
O	14003	O	7.68
I	13318	I	7.31
N	12666	N	6.95
S	11450	S	6.28
R	10977	R	6.02
H	10795	H	5.92
D	7874	D	4.32
L	7253	L	3.98
U	5246	U	2.88
C	4943	C	2.71
M	4761	M	2.61
F	4200	F	2.30
Y	3853	Y	2.11
W	3819	W	2.09
G	3693	G	2.03
P	3316	P	1.82
B	2715	B	1.49
V	2019	V	1.11
K	1257	K	0.69
X	315	X	0.17
Q	205	Q	0.11
J	188	J	0.10
Z	128	Z	0.07

Image Attribution

English Letter Frequency (based on a sample of 40,000 words)

<https://pi.math.cornell.edu/~mec/2003-2004/cryptography/subs/frequencies.html>

Improvements...

- Problem: Key Sounds Are Independent
 - Capture Sound Naturally Typing
 - Create a Map Considering Previous Key Press (Difficult)
 - Take Typing Flow Into Account
 - Need Multiple Users!
 - Method Variation: Home Row
- Environment
 - Clean Noise On-the-Fly



Real World Attack Vectors

This Isn't New...



Apple co-founder Steve Wozniak used a toy whistle from Cap'n Crunch cereal to hack AT&T and get free phone calls.

Using Single Keyboard...



Image Attribution

<https://www.amazon.com/DC-English-Computer-Keyboard-Oversized/dp/B002WP0UAG>

What About This...



Image Attribution
First National Bank
<https://www.firstnationalatm.com/what-is-an-atm-machine/>

Multiple Keyboards



Image Attribution

https://steelseries.com/en-lt/gaming-keyboards?refine=c_keyboardSize%3DMini+%2860%25%29

<https://www.homedepot.com/p/BIBIVIS-Large-Print-Keyboard-with-7-Interchangeable-Backlight-Colors-USB-Wired-Computer-Keyboards-for-Windows-PC-Laptop-AN37-L00/334261580>

They Looking For Every Edge...

**YOU'RE
A
TARGET**



Privacy, Security, And Protecting Yourself

2 Factor Authentication



(Strong) Passwords You Don't Know



KeePassXC



1Password



bitwarden

Different Passwords Per Platform



Change Password Frequently



Image Attribution

<https://www.greengeeks.com/blog/tips-for-change-your-password-day/>

Intentionally Difficult...

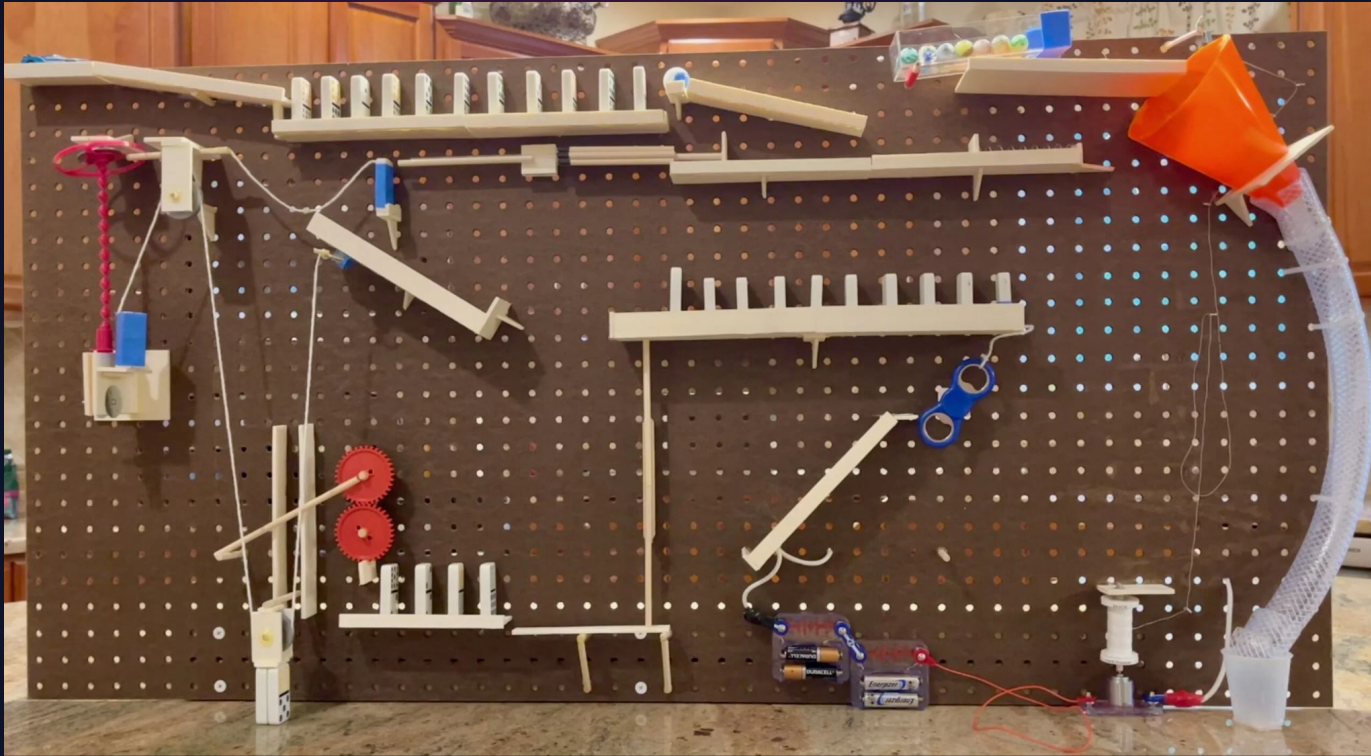


Image Attribution

<https://www.whimsicalwidgets.com/our-first-rube-goldberg-machine/>

Seems Innocent...

TECH

Apple acquires Israeli startup Q.ai

PUBLISHED THU, JAN 29 2026 • 1:41 PM EST



Kif Leswing
@KIFLESWING

Using a wearable to interpret facial skin micromovements

Abstract

Systems, methods, and non-transitory computer readable media including instructions for interpreting facial skin micromovements are disclosed. An example head mountable system includes a housing configured to be worn on a head of a wearer. The head mountable system also includes at least one detector integrated with the housing and configured to receive light reflections from a facial region of the head and to output associated reflection signals. The head mountable system also includes at least one microphone associated with the housing and configured to capture sounds produced by the wearer and to output associated audio signals. The head mountable system also includes and at least one processor in the housing, configured to use both the reflection signals and the audio signals to generate output that corresponds with words articulated by the wearer.

US20240119938A1

United States

Download PDF

Find Prior Art

Inventor: Yonatan Wexler, Avi BAR

Current Assignee: Q Cue Ltd

Worldwide applications

Image Attribution

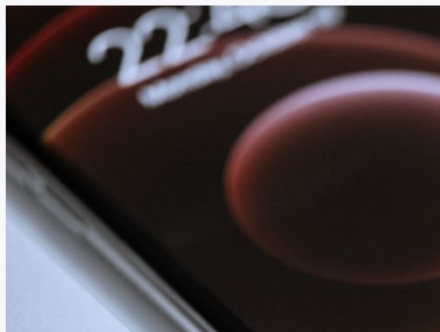
<https://www.cnbc.com/2026/01/29/apple-acquires-israeli-startup-qai-.html>

Think I Am Paranoid...

News Briefs

AI

Apple Acquires Q.ai Startup for "Silent" Voice Input



ai atlas > Tech > Services & Software > AI

Apple's New Lip-Reading Tech Is a Weird Signal of Where We're Heading

Commentary: The purchase of a startup called Q.ai isn't a blip on the radar. The way we use our wearables could be changing in very weird new ways.

Image Attribution

<https://www.cnet.com/tech/services-and-software/apple-qai-acquisition-lip-reading-tech-wearables/>

<https://www.privacyguides.org/news/2026/01/30/apple-acquires-q-ai-startup-for-silent-voice-input/>

Resources

Resources

All Materials (Slides, Code, Instructions, etc):

github.com/davidvonthenen/2026-devovx-france-keystroke

Topics:

- Research: [A Practical Deep Learning-Based Acoustic Side Channel Attack on Keyboards](#)
- Original Datasets:
 - [Single Keyboard RAW Sounds](#)
 - [Multiple Keyboard RAW Sounds](#)

Additional Resources:

- Disrupting AI and LLMs: [Devovx France 2025 - Explaining the Unexplainable: Python Tools for AI Transparency](#)
- Sarcasm Detection: [AIDev EU 2025: Sarcastically Speaking](#)



Thank You!



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